Cardiology Journal Club · Week 12

Does a smartwatch catch atrial fibrillation worth catching?

A critical read of the HEARTBEAT screening trial — and what its endpoints leave out.

The paper claims a wearable doubled the detection of silent atrial fibrillation.

Okafor et al. (2026) · N. Engl. J. Cardiol. 41:1102 · "HEARTBEAT"

> *Among 42,000 adults aged 65 and older, smartwatch-based screening identified new atrial fibrillation in 5.1% of participants versus 2.6% under usual care over twelve months.*

The headline is a doubling of detection — 5.1% versus 2.6%.

What we must do.

Decide whether *detecting* more AF means *preventing* more strokes.

The question this club is here to answer: detection or outcomes?

Finding more atrial fibrillation is only useful if treating it prevents the strokes AF causes. The trial reports detection as its primary endpoint and strokes as secondary. That ordering is the whole debate — so we read it closely.

Methods: a large, well-randomized pragmatic trial — with one soft endpoint.

42,000 community-dwelling adults were randomized to a screening smartwatch or usual care and followed for twelve months. Randomization and follow-up were rigorous. But the primary endpoint is a *diagnosis*, not a clinical event — and diagnoses are easy to inflate by simply looking harder.

- Randomization and retention are genuinely strong.
- Primary endpoint is detection, not stroke or death.

HEARTBEAT · 12-month results as reported

What the trial found, primary and secondary endpoints side by side.

n = 42,000, intention-to-treat, 12-month follow-up.

5.1%	2.6%	−0.3%	11%
AF DETECTED (SCREENED)	AF DETECTED (USUAL CARE)	STROKE RATE (NOT SIGNIFICANT)	FALSE-POSITIVE ALERTS

The detection doubled — the strokes did not budge.

The screened arm caught twice as much AF, but the stroke difference (−0.3%) did not reach significance. The trial detected a lot of low-burden, short-duration AF that may never have caused a clot. More diagnosis did not become less disease.

- Stroke reduction was the question; the trial was underpowered for it.
- Much of the extra AF was brief and low-risk.

The cost of looking harder: false positives and the cascade they trigger.

Eleven percent of screened participants got an alert that turned out to be noise. Each false positive can mean a cardiology referral, a monitor, sometimes anticoagulation — which carries its own bleeding risk. The trial counts the benefits of detection but not the harms of over-detection.

- False positives drive downstream tests and anxiety.
- Anticoagulating low-risk AF can do net harm.

Whether screening helps depends on who you screen.

High AF risk · Low bleeding risk.

Screening likely net-beneficial — treat what you find.

Low AF risk · Low bleeding risk.

Detection without benefit — incidentalomas.

High AF risk · High bleeding risk.

Diagnosis useful, but treatment is a hard call.

High AF risk · Low bleeding risk · clinically indicated.

The population the trial should have enriched for.

What we would have designed differently.

We would power for stroke, not detection; enrich for high-risk patients where treatment clearly helps; and report a harms ledger alongside the detection count. As published, HEARTBEAT proves the watch works as a sensor — not that screening works as medicine.

- Power for the clinical event, not the surrogate.
- Pre-specify and report the over-detection harms.

The verdict: a strong sensor study, a weak screening study.

The technology detects AF reliably — that much is convincing. But the leap from "detects AF" to "should screen the population" is unsupported by these endpoints. We file this as hypothesis-generating, awaiting a stroke-powered trial.

- Convincing on detection accuracy.
- Unconvincing on clinical benefit — verdict: not yet.

Discussant · Dr. Lena Fischer · next week: the stroke-powered follow-up

Detecting more disease is not the same as preventing more harm.